

Primal–Dual Curvature-Matching (PDCM)

Methodological form vs. practical concessions

The algorithm as coded in `recipe/nemo_gym/pdcm_sampler.py` + `domain_proportion_sampler.py` (run `pdcm-mix4-ifsplitt-floor15-ts-clamp02`). Part I is the principled method; Part II lists the pragmatic substitutions the deployed controller makes on top of it, each keyed to the methodological element it modifies. Notation: $\tilde{p}$ = applied proportions, $\hat{g}$ = curvature, $\hat{U}$ = observed utility, $\bar{\cdot}$ = window mean.

Part I Methodological form

Setting & objective. At each step t the controller allocates the batch over K domains via proportions $\tilde{p}_t$, $\sum_k \tilde{p}_{t,k} = 1$ (an outer loop — it re-proportions data, never touches a model gradient). It maximizes cumulative policy improvement subject to a *lag constraint*: no domain’s mean per-step improvement trails its uniform-allocation ($1/K$) counterfactual by more than a budget ε .

Signal (first order). Each domain reports a per-step improvement rate $\hat{U}_{t,k}$ — a *first-order* learning velocity (how fast performance rises this step).

Driver (second order). The allocator is steered by the *marginal utility* $\hat{g}_{t,k} \approx \partial \hat{U}_k / \partial \tilde{p}_k$ — the *second-order* quantity, the slope of the improvement rate with respect to allocation. It is the local linear-regression slope of $\hat{U}$ on $\tilde{p}$ over a length- W window; the canonical estimator is ridge least squares:

$$\hat{g}_k = \frac{\sum_{\tau} (\tilde{p}_{\tau,k} - \bar{p}_k)(\hat{U}_{\tau,k} - \bar{U}_k)}{\sum_{\tau} (\tilde{p}_{\tau,k} - \bar{p}_k)^2 + \nu} \quad (\nu : \text{ridge, division-by-zero guard}).$$

Principle: push allocation toward domains whose marginal utility $\hat{g}$ is high (learning there is still accelerating), and away from those where it has flattened.

Primal — entropic mirror descent. Exponentiated-gradient ascent of the dual-modified marginal utility on the simplex:

$$p_{t+1} = \text{softmax}(\log p_t + \eta_p \hat{g}(1 + \lambda)).$$

Dual — the lag constraint. A per-domain multiplier $\lambda_k \geq 0$ rises (projected ascent, tangent-line bound of the unobserved uniform baseline) when a domain is under-allocated ($\frac{1}{K} - \tilde{p}_k > 0$) and still improvable ($\hat{g}_k > 0$) beyond the budget, then amplifies that domain’s $\hat{g}$ in the primal:

$$\lambda_{t+1,k} = \max(0, \lambda_{t,k} + \eta_d [\hat{g}_k (\frac{1}{K} - \tilde{p}_{t,k}) - \varepsilon]).$$

Excitation. $\hat{g}$ is only identifiable if $\tilde{p}$ actually varies, so the executed proportions carry annealed persistent-excitation dither: $\tilde{p}_t = (1 - \gamma_t)p_t + \gamma_t \zeta$, $\zeta \sim \text{Dir}(\mathbf{1}_K)$, γ_t decaying $\gamma_{\text{init}} \rightarrow \gamma_{\text{floor}}$.

Part II Practical concessions

The deployed controller keeps the skeleton above but makes the substitutions below. Each is an engineering choice or a fix for an observed pathology — *not* part of the method — and the mix4 runs showed each was needed.

1. Theil–Sen curvature (replaces the ridge-OLS estimator of $\hat{g}$). The method only asks for the second-order slope; the *choice* of estimator is ours. OLS’s squared loss lets a single noisy $\hat{U}$ tick or one high-leverage $\tilde{p}$ swing dominate $\hat{g}$. We use the outlier-robust median of pairwise slopes ($\sim 29\%$ breakdown in both $\hat{U}$ and $\tilde{p}$):

$$\hat{g}_k = \text{median}_{i < j} \frac{\hat{U}_{j,k} - \hat{U}_{i,k}}{\tilde{p}_{j,k} - \tilde{p}_{i,k}}, \quad |\Delta \tilde{p}| > \text{ts_eps} \quad (\text{opt. shrink} \times \frac{\text{Var}_k}{\text{Var}_k + \text{ts_ridge}}).$$

2. Hard curvature cap (adds a bound on $\hat{g}$; not in the method). $\hat{g}_k \leftarrow \text{clip}(\hat{g}_k, -g_{\text{clip}}, +g_{\text{clip}})$. Numerical backstop: a collapsed-variance window can otherwise blow the regression slope up (a v1 spike of $\hat{g} \approx -9 \times 10^4$) and detonate the softmax.

3. Log-odds anchor clamp (modifies the primal recursion). $p_{t+1} = \text{softmax}(\log \max(p_t, c) + \eta_p \hat{g}(1 + \lambda))$, $c = \text{primal_clamp}$. The exact mirror-descent update is an integrator (log-odds = running sum of $\hat{g}$); a sustained (possibly phantom-) negative slope winds a domain into an unrecoverable $\log(\approx 0)$ “pit.” Clamping only the *anchor* bounds the depth — few-step recovery instead of ~ 50 — while the stored p_t may still sit below c so the floor is still reachable. ($c \ll f$.)

4. Hard per-domain floor (*replaces reliance on the dual/excitation for allocation safety*). $\tilde{p}_t = f + (1 - Kf)w$, $f = \text{prop_floor}$, on the executed proportions after dither. The methodological safety mechanism (the dual lag constraint) alone let allocation collapse — the pure-method run pinned IF at $\tilde{p} \approx 0.85$ and starved code_gen to $\approx 2/128$. The affine floor guarantees $\tilde{p}_k \geq f$ and caps the max arm at $1 - (K - 1)f$, and floors *without pinning* (every arm keeps a $(1 - Kf)^2$ share of the dither variance). *Caveat*: it *bounds* misallocation but does not *restore* the curvature signal — that remains excitation-limited.

5. Utility proxy & amplifier (*instantiates $\hat{U}$*). $\hat{U}_{t,k} = A \cdot \Delta \text{perf}_{t,k}$, $A = \text{perf_diff_amplifier} = 100$. The true improvement rate is proxied by the realized per-domain perf-diff, scaled so the small deltas move the exponentiated gradient meaningfully.

6. Per-run lag budget (*bookkeeping on the dual's ε*). The constant ε is applied as $\varepsilon s/T$ ($s = \text{steps_per_update}$, $T = \text{total_steps}$) so the cumulative slack equals ε over the whole run rather than per update.

7. Dithering demoted (*reprioritized*). Excitation is retained for identifiability but is now *secondary* to the floor for allocation safety (the floor, not the dither, is the hard rail).

Run configuration (pdcn-mix4-ifsplitt-floor15-ts-clamp02)

$K = 4$	$\eta_p = 0.1$	$\eta_d = 0.01$	$W = 10$
$\gamma_{\text{init}} = 0.20$	$\gamma_{\text{floor}} = 0.10$	$T_{\text{anneal}} = 400$	warmup = 20
$f = 0.15$	$c = 0.02$	$g_{\text{clip}} = 5$	$\nu = 0.01$
$\varepsilon = 0.1$	$A = 100$	estimator = Theil-Sen	ts_ridge = 0.01
$T = 600$	loss = GSPO	adv = GRPO	curator OFF